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Backend Software Architecture and Data Model Documentation

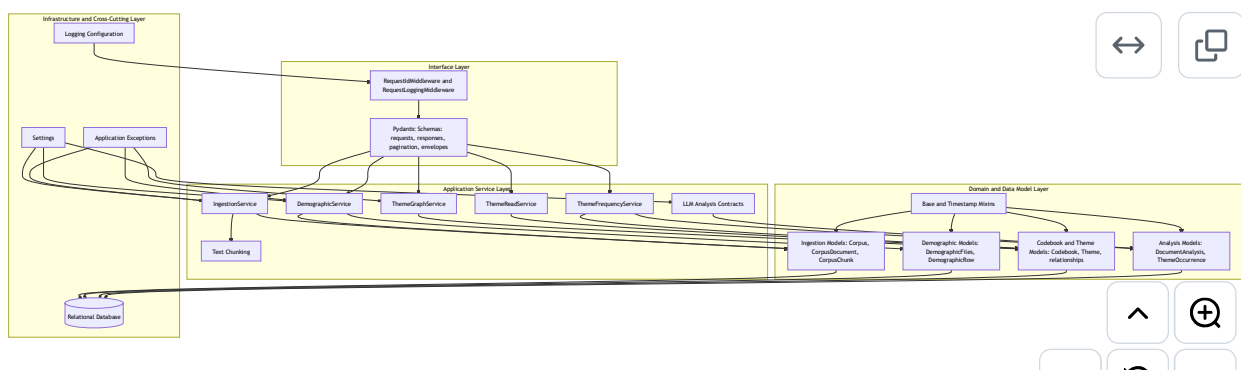
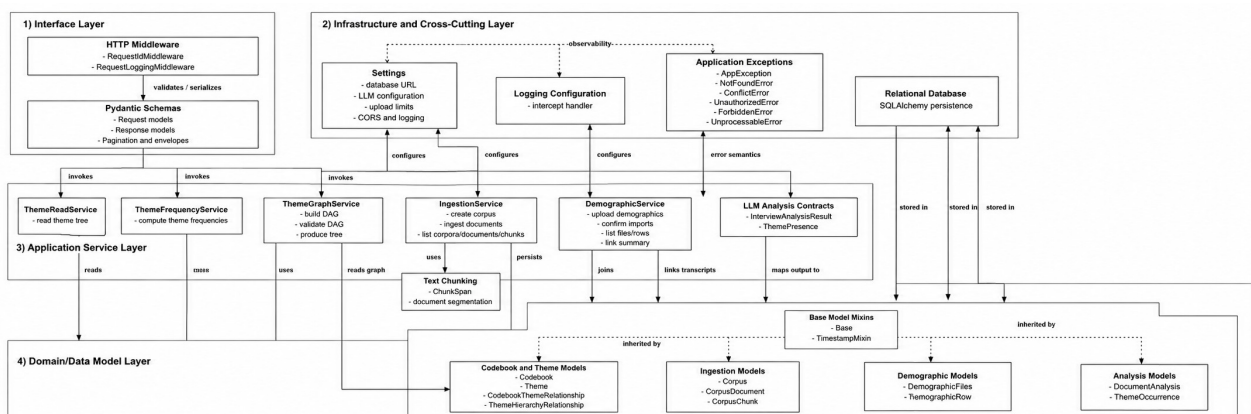
Source: generated from the `classes_BackendApp.dot` pyreverse class graph.

This document presents a non-decorative architectural abstraction of the backend. It separates runtime responsibilities from persistence structures and avoids implementation-specific visual clutter.

1. Architectural Overview

The backend is organized as a layered application. The observable structure indicates the following principal layers:

1. **Interface layer:** request/response schemas, pagination envelopes, middleware, and validation contracts.
2. **Application service layer:** procedural business operations for ingestion, demographic processing, theme graph computation, theme reading, and frequency aggregation.
3. **Domain/data model layer:** SQLAlchemy-backed persistent entities representing corpora, documents, demographic files, codebooks, themes, analyses, and theme occurrences.
4. **Infrastructure and cross-cutting layer:** configuration, logging, and domain-specific exception classes.





2. Main Service Responsibilities

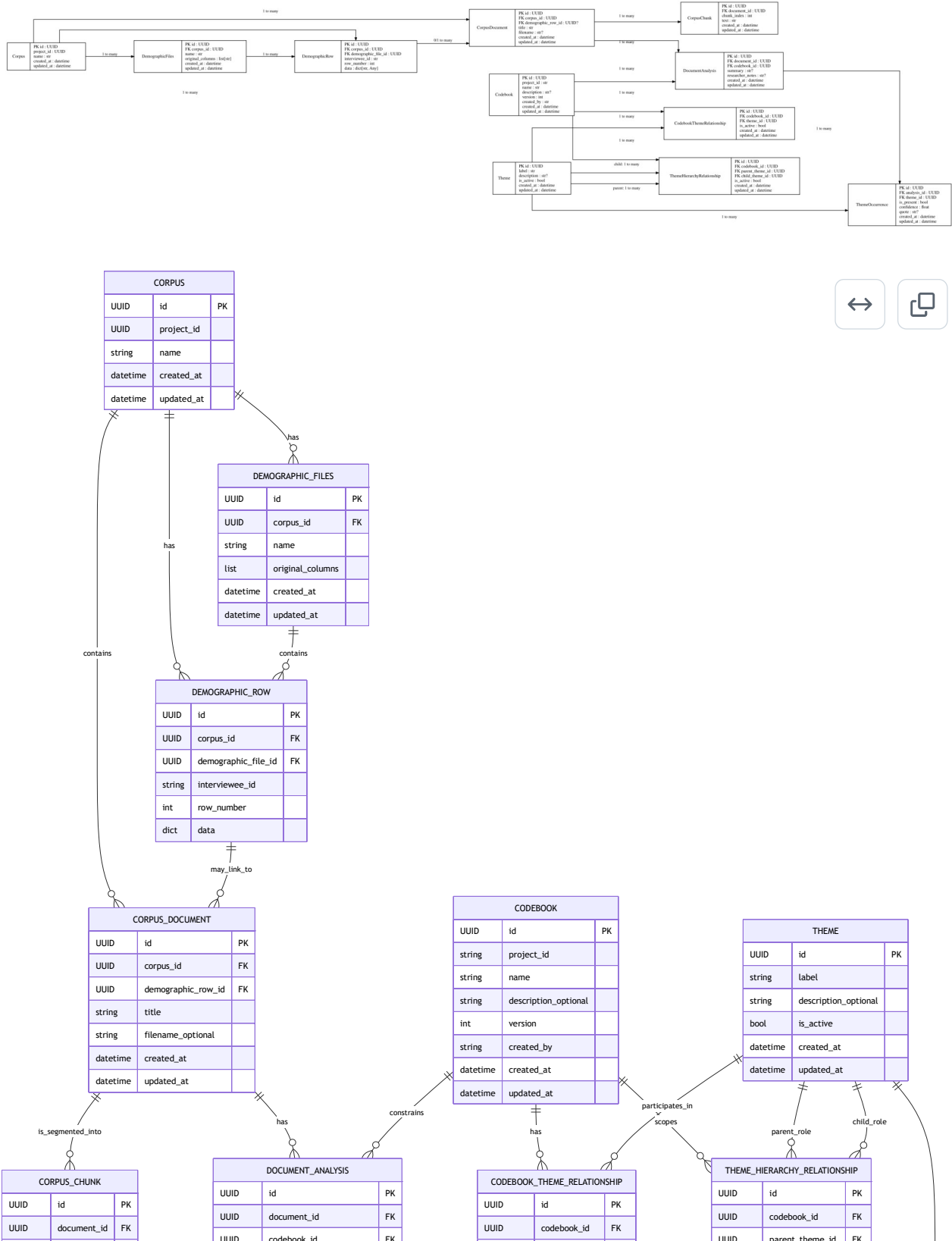
Service or component	Scientific role	Primary data dependency
IngestionService	Creates corpora, ingests documents, segments text into chunks, and lists corpus artifacts.	Corpus , CorpusDocument , CorpusChunk
DemographicService	Imports demographic CSV-like data, confirms temporary uploads, lists demographic records, and links demographic rows to transcripts.	DemographicFiles , DemographicRow , CorpusDocument
ThemeGraphService	Constructs and validates a directed acyclic graph of themes within a codebook.	Theme , ThemeHierarchyRelationship , Codebook
ThemeReadService	Provides a readable tree representation of the theme hierarchy.	Theme , ThemeHierarchyRelationship
ThemeFrequencyService	Computes aggregate theme occurrence statistics across analyses.	ThemeOccurrence , Theme , DocumentAnalysis
Text chunking components	Convert raw document text into indexed spans suitable for storage and analysis.	ChunkSpan , CorpusChunk
.....	Represent structured model output before	InterviewAnalysisResult , ThemePresence ,

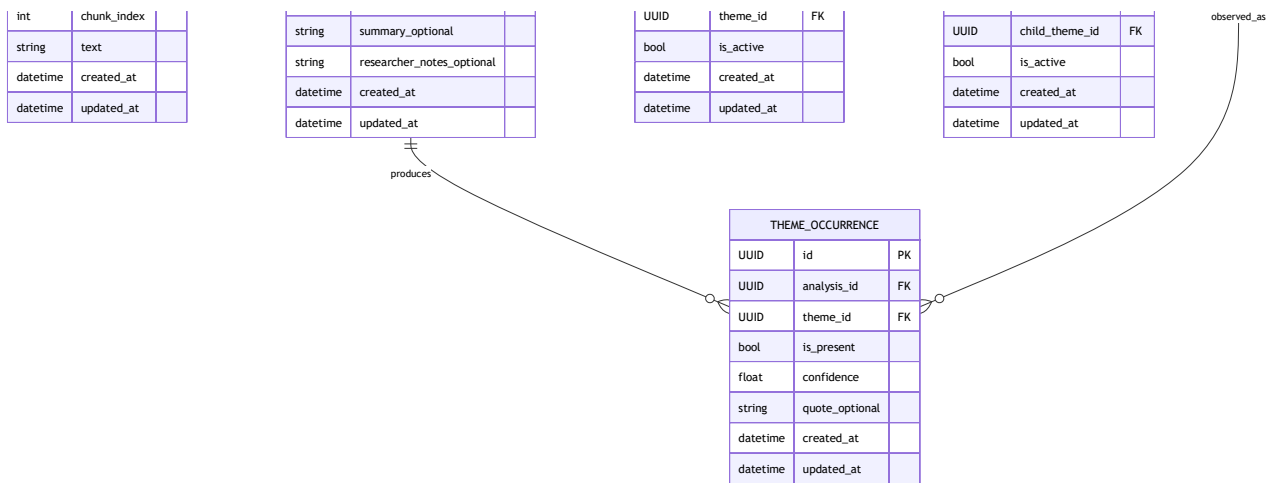
LLM analysis contracts

persistence into
analysis tables.DocumentAnalysis ,
ThemeOccurrence

3. Persistent Data Model

The following model is a conceptual ERM derived from the class graph and the foreign-key-like attributes visible in the model classes.





4. Entity Catalogue

4.1 Corpus and document ingestion

Entity	Description	Key attributes
Corpus	Logical collection of source documents within a project.	id , project_id , name
CorpusDocument	A single ingested document or transcript. It may be linked to one demographic row.	id , corpus_id , demographic_row_id , title , filename
CorpusChunk	Indexed segment of a document used for processing or analysis.	id , document_id , chunk_index , text

4.2 Demographic data

Entity	Description	Key attributes
DemographicFiles	Imported demographic file metadata and original column structure.	id , corpus_id , name , original_columns
DemographicRow	One normalized row of imported demographic data.	id , corpus_id , demographic_file_id , interviewee_id , row_number , data

4.3 Codebook and theme graph

Entity	Description	Key attributes
Codebook	Versioned coding framework associated with a project.	id , project_id , name , description , version , created_by
Theme	Atomic code or theme used in qualitative analysis.	id , label , description , is_active
CodebookThemeRelationship	Association table mapping themes into codebooks.	id , codebook_id , theme_id , is_active
ThemeHierarchyRelationship	Directed edge between parent and child themes within a codebook.	id , codebook_id , parent_theme_id , child_theme_id , is_active

4.4 Analysis

Entity	Description	Key attributes
DocumentAnalysis	Analysis result for a document under a specific codebook.	id , document_id , codebook_id , summary , researcher_notes
ThemeOccurrence	Observation of a theme within a document analysis, including evidence and confidence.	id , analysis_id , theme_id , is_present , confidence , quote

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